

TrES-3b

R-2788

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130715 · created 2026-09-04T00:44:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456488.7162 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.055
Transit depth (Rp/Rs)^2	3.81 +/- 2.15 [%]
Orbital Inclination (inc)	86.95 +/- 2.92
Airmass coefficient 1 (a1)	0.529 +/- 0.03
Airmass coefficient 2 (a2)	0.63 +/- 0.27
Scatter in the residuals of the lightcurve fit is	5.7 %
Best Comparison Star	#1 - [556, 414]
Optimal Aperture	3.66
Optimal Annulus	15.04
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.055 vs 0.16309 (deviation 0.55σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 2.02σ)
Mid-time O–C	-2.7 min
Depth SNR	3.4
Residual scatter	5.7 %

Light curve

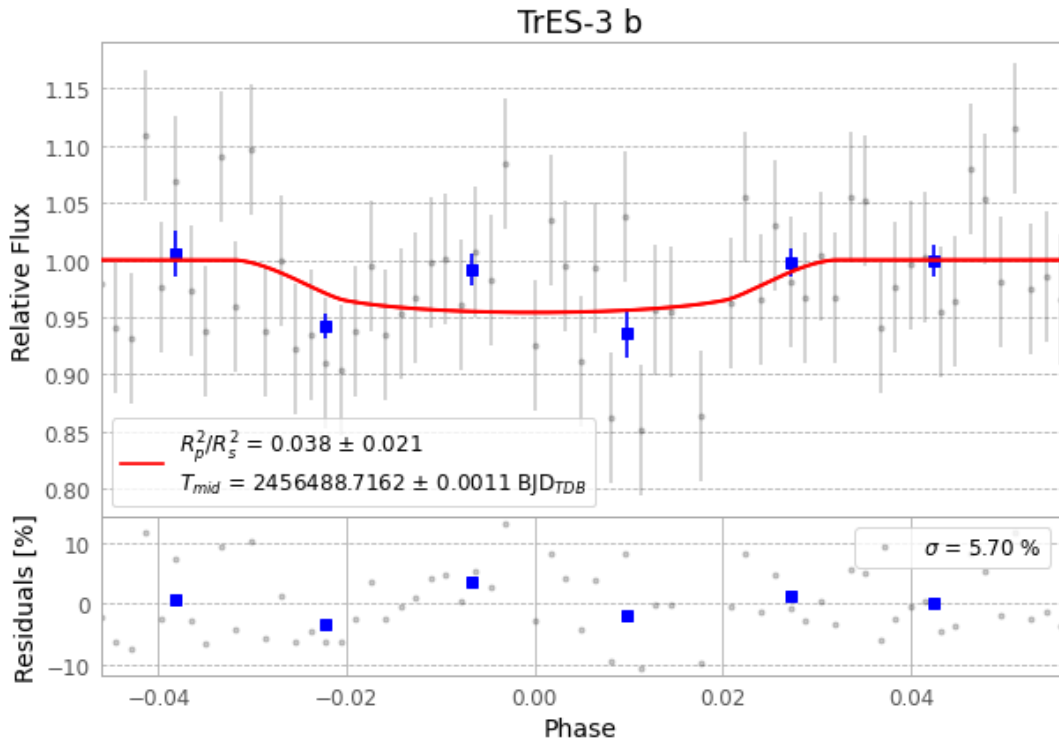


Figure 1 — FinalLightCurve_TrES-3 b_2013-07-14.png (SHA-256 49f70d8433d4f072...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c966dd01705c2a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3130715033912.FITS → TRES-3130715065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130715.log (e60e2165838a...), FinalLightCurve_TrES-3 b_2013-07-14.png (49f70d8433d4...), FinalLightCurve_TrES-3 b_2013-07-14.csv (3afb98f1e1d1...)

Compute resources

Wall time 191.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:44Z.